MATH50003 Numerical Analysis

VI.1 Chebyshev Polynomials

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Course content

I. Calculus on a Computer
   - Integration, differentiation, root finding

II. Representing Numbers
   - Floating point numbers, bounding errors, interval arithmetic

III. Numerical Linear Algebra
   - Structured matrices, LU & QR factorisations, least squares

IV. Linear Algebra Applications
   - Data regression, singular value decomposition, matrix compression

V. Numerical Fourier series
   - Fourier expansions and transforms, convolutions

VI. Orthogonal Polynomials
   - Chebyshev polynomials

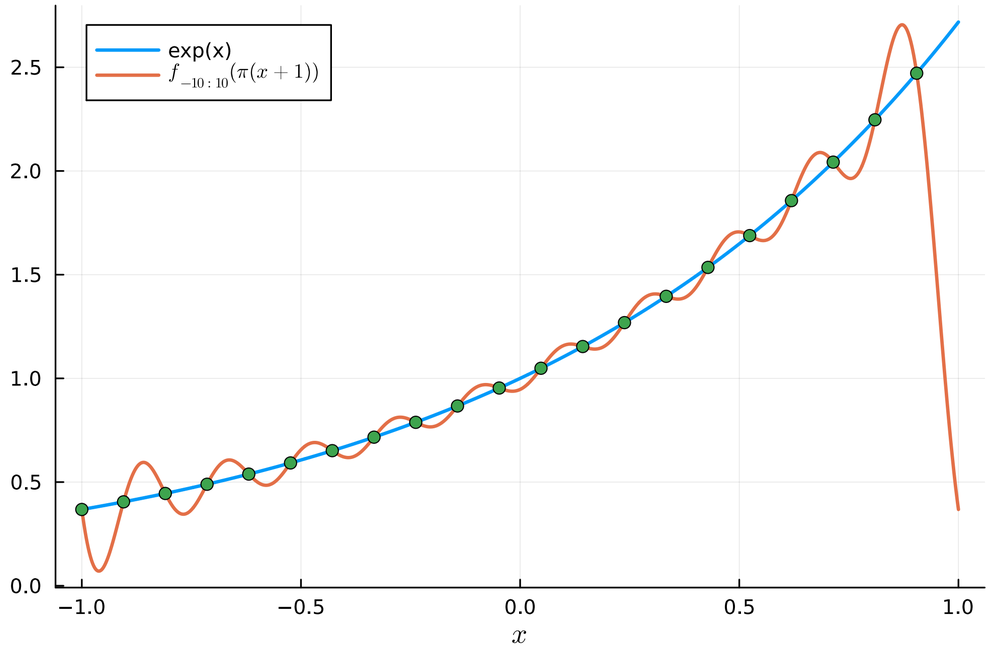
Chapter VI

Orthogonal Polynomials

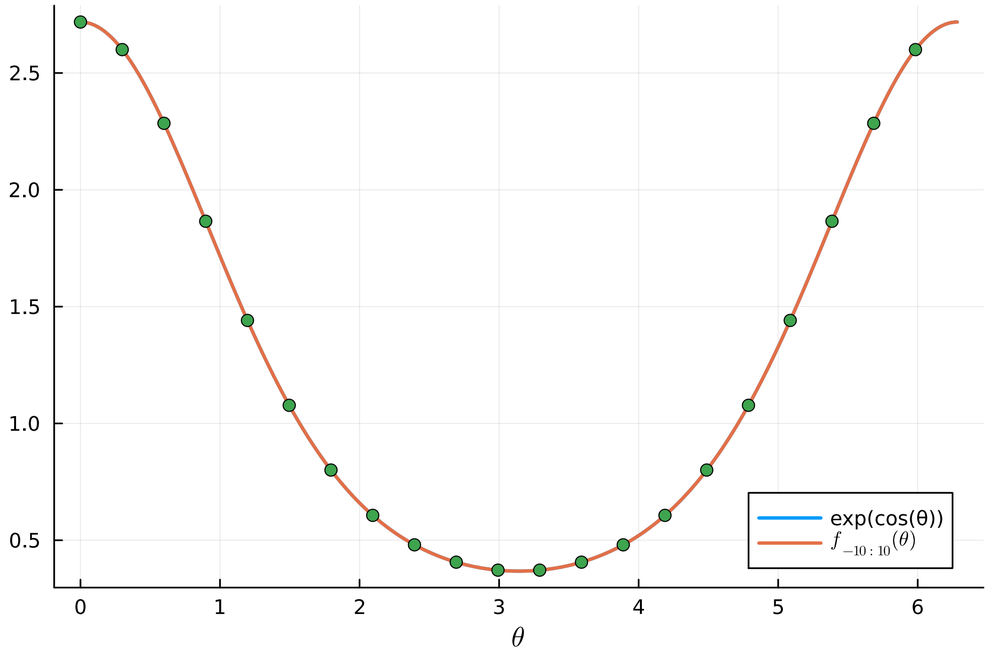
1. Chebyshev polynomials as Fourier in-disguise

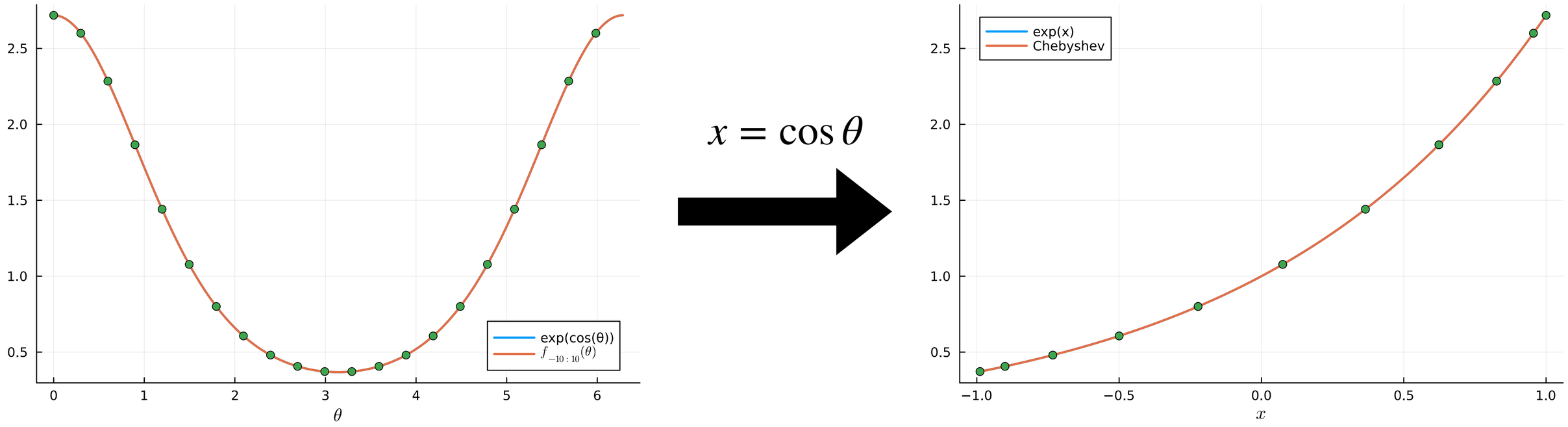
How to approximate non-periodic functions?

Use change-of-variables to map to Fourier.



$$x = \cos\theta$$



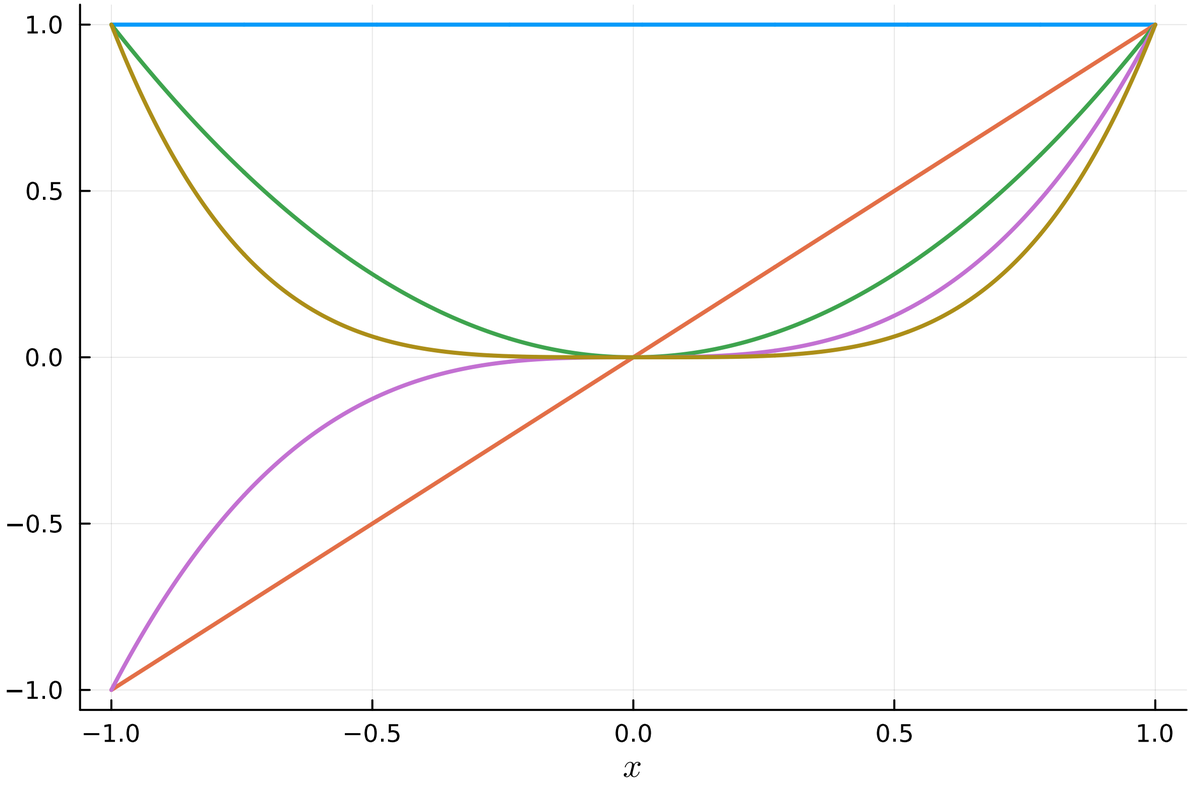

2.5
2.0
1.5
1.0
0.5
0
1
2
3
4
5
6
θ
exp(cos(θ))
$f_{-10:10}(\theta)$
$x = \cos\theta$
exp(x)
Chebyshev
−1.0
−0.5
0.0
0.5
1.0
x

VI.1.1 Chebyshev polynomials are orthogonal polynomials

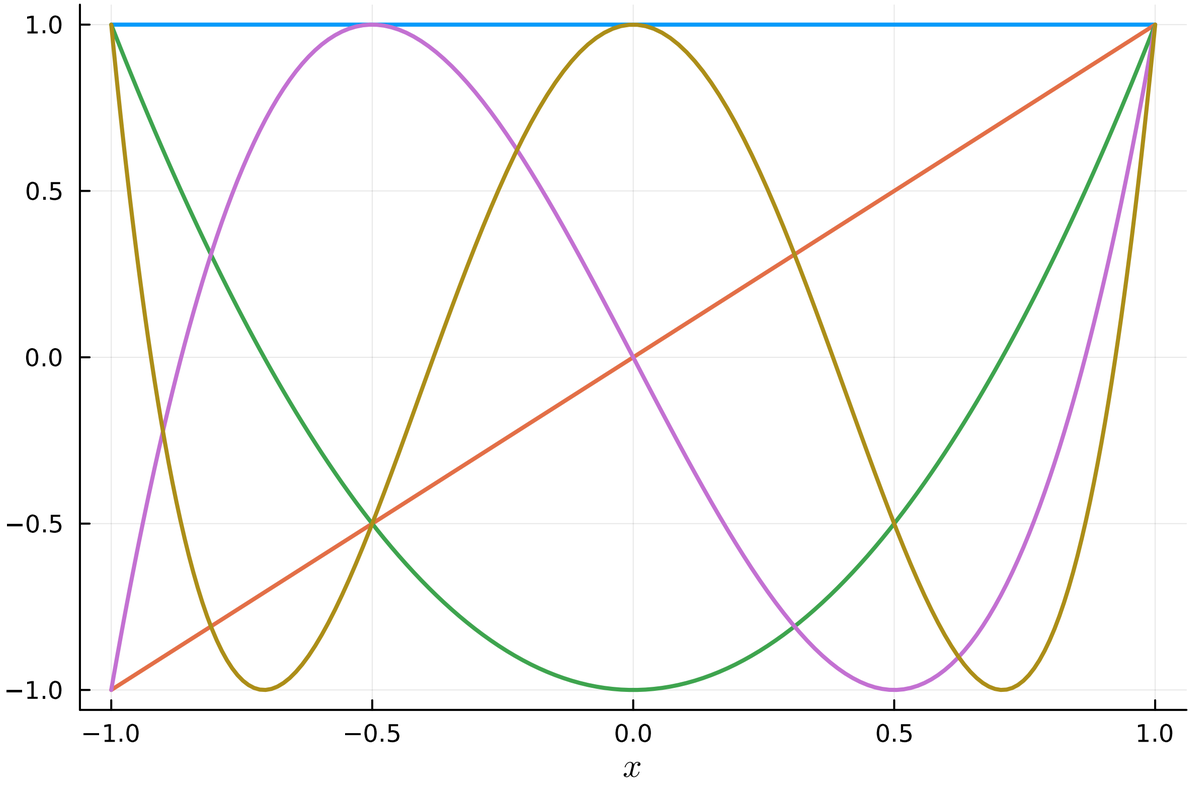
They are (1) polynomials in x and (2) orthogonal wrt $1/\sqrt{1-x^2}$

Lemma 10 (Chebyshev 3-term recurrence).

$$xT_0(x) = T_1(x)$$
$$xT_n(x) = \frac{T_{n-1}(x) + T_{n+1}(x)}{2}$$



Monomials



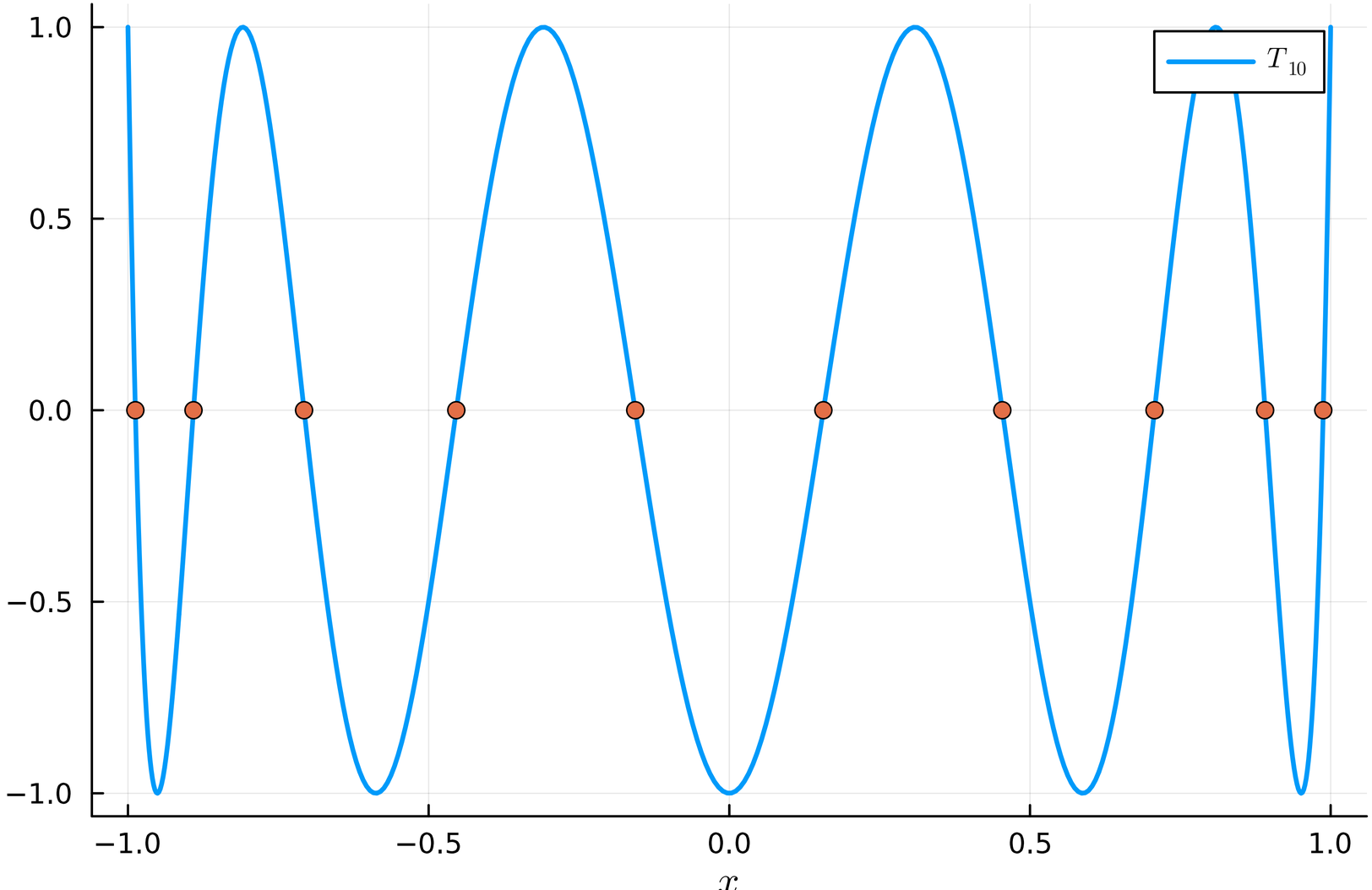
Chebyshev

Theorem 17 (Chebyshev polynomials, 1st kind)**.** *$T_n(x)$ are orthogonal polynomials with respect to the inner product*

$$\int_{-1}^{1} \frac{f(x)g(x)}{\sqrt{1-x^2}} \mathrm{d}x.$$

VI.1.2 Interpolation at roots of Chebyshev polynomials

A stable polynomial basis and grid for interpolation



Theorem 18 (inverse Chebyshev Vandermonde)**.** *We have*

$$V^{-1} = \frac{1}{n} \begin{bmatrix} 1 & & & \\ & 2 & & \\ & & \ddots & \\ & & & 2 \end{bmatrix} V^{\top}.$$